RAJALAKSHMI ENGINEERING COLLEGE

Lokesh S 251801161-2025-AIDS

L2

Started on	Thursday, 24 September 2026, 11:03 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 8:45 PM
Time taken	13 days 9 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int array[n],array2=0;
    for(int i=0;i<n;i++)
    {
        scanf("%d",&array[i]);
    }
    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(array[i]==array[j])
            {
                array2=array[j];
            }
        }
    }
    printf("%d ",array2);
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lokesh S 251801161-2025-AIDS

L2

Started on	Thursday, 24 September 2026, 11:04 AM
State	Finished
Completed on	Saturday, 10 October 2026, 9:01 PM
Time taken	16 days 9 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int array[n],array2=0;
    for(int i=0;i<n;i++)
    {
        scanf("%d",&array[i]);
    }
    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(array[i]==array[j])
            {
                array2=array[j];
            }
        }
    }
    printf("%d ",array2);
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lokesh S 251801161-2025-AIDS

L2

Started on	Thursday, 24 September 2026, 11:04 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:20 AM
Time taken	13 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
int p;
scanf("%d",&p);
int n,m;
int i;
scanf("%d",&n);
int array[n];
for(i=0;i<n;i++)
{
scanf("%d",&array[i]);
}
scanf("%d",&m);
int array2[m];
for(i=0;i<m;i++)
{
scanf("%d",&array2[i]);
}
for(i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
```

```
    {
        if(array[i]==array2[j])
        {
            printf("%d ",array[i]);
            break;
        }
    }
}
return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lokesh S 251801161-2025-AIDS

L2

Started on	Thursday, 8 October 2026, 10:20 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:20 AM
Time taken	13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
int p;
scanf("%d",&p);
int n,m;
int i;
scanf("%d",&n);
int array[n];
for(i=0;i<n;i++)
{
scanf("%d",&array[i]);
}
scanf("%d",&m);
int array2[m];
for(i=0;i<m;i++)
{
scanf("%d",&array2[i]);
}
for(i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
```

```
    {
        if(array[i]==array2[j])
        {
            printf("%d ",array[i]);
            break;
        }
    }
}
return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lokesh S 251801161-2025-AIDS

L2

Started on	Saturday, 10 October 2026, 9:01 PM
State	Finished
Completed on	Saturday, 10 October 2026, 9:02 PM
Time taken	47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n, k;
    scanf("%d", &n);

    int A[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &A[i]);
    }

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n)
    {
        if (i == j)
        {
            j++;
        }
        else if (A[j] - A[i] == k)
        {
            printf("1");
            return 0;
        }
        else if (A[j] - A[i] < k)
        {
            j++;
        }
        else
        {
            i++;
        }
    }

```

```
40      printf("0");
41
42      return 0;
43  }
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lokesh S 251801161-2025-AIDS

L2

Started on	Saturday, 10 October 2026, 9:02 PM
State	Finished
Completed on	Saturday, 10 October 2026, 9:03 PM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n, k;
    scanf("%d", &n);

    int A[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &A[i]);
    }

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n)
    {
        if (i == j)
            j++;
        else if (A[j] - A[i] == k)
        {
            printf("1");
            return 0;
        }
        else if (A[j] - A[i] < k)
            j++;
        else
            i++;
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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